

Deep learning theory: lecture 6

Recall: error decomposition

$$\underbrace{R(f_{\hat{\theta}}) - \inf_{f: X \rightarrow Y} R(f)}_{\text{population risk}} = \underbrace{R(f_{\hat{\theta}}) - \inf_{\theta \in \Theta} R(f_{\theta})}_{\text{estimation err.}} + \underbrace{\inf_{\theta \in \Theta} R(f_{\theta}) - \inf_{f: X \rightarrow Y} R(f)}_{\text{approximation err.}}$$

ANALYSIS (F)

typically $\xrightarrow[n \rightarrow \infty]{|\Theta| \rightarrow \infty} 0$ typically $\xrightarrow{|\Theta| \rightarrow \infty} 0$

we can further decompose the estimation error as:

$$\underbrace{R(f_{\hat{\theta}}) - R(f_{\theta^*})}_{\text{estimation}} = \underbrace{R(f_{\hat{\theta}}) - \hat{R}(f_{\hat{\theta}})}_{\text{gen / concentr.}} + \underbrace{\hat{R}(f_{\hat{\theta}}) - \hat{R}(f_{\theta^*})}_{\text{empirical optimiz}} + \underbrace{\hat{R}(f_{\theta^*}) - R(f_{\theta^*})}_{\text{gen}}$$

STA ($\mathcal{D}_n$) OPT ($\mathcal{F} + \mathcal{F}$)

$\Rightarrow$ We aim to bound terms of the form

$$\hat{R}(f_{\theta}) - R(f_{\theta})$$

Example (linear regression):

$$\begin{aligned} \sup_{\|\theta\| \leq D} \hat{R}(\theta) - R(\theta) &= \sup_{\|\theta\| \leq D} \frac{1}{n} \sum (x_j \theta - y_j)^2 - \mathbb{E}_p (x \theta - y)^2 \\ &= \sup_{\|\theta\| \leq D} \theta^T \left(\frac{1}{n} \sum x_j x_j^T - \mathbb{E}(xx^T) \right) \theta \end{aligned}$$

$$\begin{aligned}
& -2\theta^T \left(\frac{1}{n} \sum y_i x_i - \mathbb{E}(yx) \right) \\
& + \left(\frac{1}{n} \sum y_i^2 - \mathbb{E}(y^2) \right) \\
& \leq D^2 \left\| \frac{1}{n} \sum x_i x_i^T - \mathbb{E}(xx^T) \right\|_{\text{op}} \\
& \quad + 2D \left\| \frac{1}{n} \sum y_i x_i - \mathbb{E}(yx) \right\|_2 + \left| \frac{1}{n} \sum y_i^2 - \mathbb{E}(y^2) \right|
\end{aligned}$$

The last term decays as $O(\frac{1}{\sqrt{n}})$ by martingale inequality,

$$\mathbb{P}\left(\left| \frac{1}{n} \sum y_i - \mathbb{E}(y) \right| > t\right) \leq \frac{\text{Var}\left(\frac{1}{n} \sum y_i^2 - \mathbb{E} y^2\right)}{t^2} = \frac{\text{Var} y^2}{t^2 \cdot n}$$

and it is natural to expect that other terms do, too.

$$\Rightarrow \mathbb{P}\left(\sup_{\|\theta\| \leq D} \hat{R}_n(\theta) - R(\theta) > t\right) \leq \kappa \frac{1}{t^2 n}$$

Choosing $t_n = \sqrt{\kappa/\delta n}$ for any $\delta \in (0, 1)$.

$$\mathbb{P}\left(\sup_{\|\theta\| \leq D} \hat{R}_n(\theta) - R(\theta) > \sqrt{\frac{\kappa/\delta}{n}}\right) \leq \delta$$

Spitzer: This is "always" the case: one expects that with probability at least $1-\delta$

$$\sup_{\theta \in \Theta} \hat{R}_n(\theta) - R(\theta) \leq \frac{\text{"Model complexity"}}{\sqrt{n} \sqrt{\delta}}$$

In general, to make this term small one needs to have

$n \gg$ "Model complexity" (n) $\rightarrow$ # of parameters.

Research has focused on defining measures of complexity that "scale well" in eg. the number of parameters of the model

Next, we give an example of such a complexity measure:

Def (Rademacher complexity): for $\varepsilon \sim \text{i.i.d.}(\{-1, 1\})^n$ define the RC of a function class $\mathcal{F}$ on a set S

$$\text{Rad}(\mathcal{F}, S) = \mathbb{E}_{\varepsilon} \left(\sup_{f \in \mathcal{F}} \langle \varepsilon, f \rangle \right) = \mathbb{E}_{\varepsilon} \sup_{f \in \mathcal{F}} \sum_{j=1}^n \varepsilon_j f(x_j)$$

and the (averaged) Rademacher complexity

$$\text{Rad}_n(\mathcal{F}) = \mathbb{E}_{\mathcal{D}_n} \left(\frac{1}{n} \text{Rad}(\mathcal{F}, S) \right)$$

This measure of complexity quantifies the ability of a model to fit $\{-1, 1\}$ random noise:

Example: Consider $\mathcal{F}_1 = \{x \mapsto \alpha_1 : \alpha_1 \in [-D, D]\}$ $n=2$

$$\mathcal{F}_2 = \{x \mapsto \alpha_1 \mathbb{1}_{x \leq 0} + \alpha_2 \mathbb{1}_{x > 0} : \alpha_1, \alpha_2 \in [-D, D]\} \quad (x, y) \sim \mathcal{N}(0, 1) \times \text{Ber}(p)$$

$$\mathcal{F}_3 = \{x \mapsto \alpha_1 \mathbb{1}_{x \leq \alpha_3} + \alpha_2 \mathbb{1}_{x > \alpha_3} : \alpha_1, \alpha_2 \in [-D, D], \alpha_3 \in \mathbb{R}\}$$

D

$$\text{Rad}_2(\mathcal{F}_1) = \mathbb{E}_{\varepsilon} \left(\frac{1}{2} \sup_{|\alpha_1| \leq D} \alpha_1 \varepsilon_1 + \alpha_1 \varepsilon_2 \right) = \frac{1}{2} \mathbb{E}_{\varepsilon} \left(D (\text{sign} \varepsilon_1 \cdot \varepsilon_1 + \text{sign} \varepsilon_1 \cdot \varepsilon_2) \right) =$$

$$= \frac{1}{2} D \left(\mathbb{E}_{\varepsilon} [|\varepsilon_1|] + \mathbb{E} (\text{sign} \varepsilon_1) \cdot \mathbb{E}_{\varepsilon} (\varepsilon_1) \right) = \frac{1}{2} D$$

$$\begin{aligned} \text{Rad}_2(\mathcal{F}_3) &= \mathbb{E}_{\mathcal{D}_2} \left[\mathbb{E}_{\varepsilon} \left(\frac{1}{2} \sup_{|\alpha_1|, |\alpha_2| \leq D} (\alpha_1 \mathbb{1}_{x_1 \leq \alpha_3} + \alpha_2 \mathbb{1}_{x_1 > \alpha_3}) \varepsilon_1 + (\alpha_1 \mathbb{1}_{x_2 \leq \alpha_3} + \alpha_2 \mathbb{1}_{x_2 > \alpha_3}) \varepsilon_2 \right) \right] \\ &= \frac{1}{2} \mathbb{E}_{\mathcal{D}_2} \left[\mathbb{E}_{\varepsilon} \left(\sup_{\alpha_1, \alpha_2, \alpha_3} \alpha_1 (\mathbb{1}_{x_1 \leq \alpha_3} \varepsilon_1 + \mathbb{1}_{x_2 \leq \alpha_3} \varepsilon_2) + \alpha_2 (\mathbb{1}_{x_1 > \alpha_3} \varepsilon_1 + \mathbb{1}_{x_2 > \alpha_3} \varepsilon_2) \right) \right] \end{aligned}$$

$$\begin{aligned} \alpha_1 &= D \operatorname{sign} \varepsilon_1 = \frac{D}{2} \mathbb{E}_{\mathcal{D}_2} \left[\mathbb{E}_{\varepsilon} \left(\operatorname{sign}(\varepsilon_1) \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} \varepsilon_1 + \operatorname{sign}(\varepsilon_2) \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} \varepsilon_2 + \right. \right. \\ &\quad \left. \left. \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} \operatorname{sign}(\varepsilon_1) \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} \varepsilon_1 + \operatorname{sign}(\varepsilon_2) \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} \varepsilon_2 \right) \right] \\ &+ D \operatorname{sign} \varepsilon_1 \\ &\quad \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} = \frac{D}{2} \mathbb{E}_{\mathcal{D}_2} \left(2 \cdot \mathbb{1}_{x_1 \leq \frac{x_1+x_2}{2}} + 2 \mathbb{1}_{x_2 \leq \frac{x_1+x_2}{2}} \right) = D \end{aligned}$$

$$\operatorname{Rad}_1(\mathcal{F}) = (\text{exercise})$$

Thm: Let $\mathcal{F}$ satisfy $f(x) \in [a, b]$ a.s. $\forall f \in \mathcal{F}$. Then w.p. $1-\delta$:

$$\sup_{f \in \mathcal{F}} \mathbb{E} f(x) - \frac{1}{n} \sum f(x_i) \leq \frac{2}{n} \operatorname{Rad}(\mathcal{F}_S) + 3(b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}}$$

Proof a) $\sup_{\mathcal{F}} \mathbb{E} f(x) - \frac{1}{n} \sum f(x_i) \leq \mathbb{E}^{\otimes n} \left(\sup_{\mathcal{F}} \mathbb{E}(f(x)) - \frac{1}{n} \sum f(x_i) \right) + (b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}}$

b) $\operatorname{Rad}_n(\mathcal{F}) \leq \frac{1}{n} \operatorname{Rad}(\mathcal{F}_S) + (b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}}$

The claim follows by a)+b); together with

Lemma $\mathbb{E}^{\otimes n} \left(\sup_{f \in \mathcal{F}} \mathbb{E}(f) - \frac{1}{n} \sum f \right) \leq \mathbb{E}^{\otimes n} \mathbb{E}^{\otimes n} \left(\sup_{f \in \mathcal{F}} \frac{1}{n} \sum f(x_i) - \frac{1}{n} \sum f(x_i) \right)$

$$\leq 2n \operatorname{Rad}_n(\mathcal{F})$$

$$\Rightarrow \sup_{\mathcal{F}} \mathbb{E} f(x) - \frac{1}{n} \sum f(x_i) \leq \mathbb{E}^{\otimes n} \left(\sup_{\mathcal{F}} \mathbb{E}(f(x)) - \frac{1}{n} \sum f(x_i) \right) + (b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}}$$

$$\leq \frac{2}{n} \mathbb{E}^{\otimes n} \operatorname{Rad}(\mathcal{F}_S) + (b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}}$$

$$\leq 2 \operatorname{Rad}_n(\mathcal{F}) + 3(b-a) \sqrt{\frac{\log \frac{1}{\delta}}{2n}} \quad \checkmark$$

Example (linear regression) Assume that $\|\theta\| \leq D$

$$\text{Rad}_n(\mathcal{F}) = \mathbb{E} \left(\sup_{\|\theta\| \leq D} \frac{1}{n} \sum \varepsilon_j \theta^\top x_j \right) = \frac{D}{n} \mathbb{E} (\| \sum \varepsilon_j x_j \|)$$

$$\stackrel{j}{\leq} \frac{D}{n} \sqrt{\mathbb{E} \left(\sum x_j^\top \varepsilon_j^2 x_j \right)} = \frac{D}{n} \sqrt{n \mathbb{E} (\|x\|^2)} = \frac{D \sqrt{\mathbb{E} (\|x\|^2)}}{\sqrt{n}}$$

$$\Rightarrow \hat{R}(\theta) - R(\theta) \leq \frac{D \sqrt{\mathbb{E} (\|x\|^2)} + (b-a) \log \left(\frac{1}{\delta} \right)}{\sqrt{n}} \quad \text{if } \text{supp}(x) \text{ is compact.}$$

Example (single layer neural networks):

$$f_\theta(x) = \sum_{j=1}^m a_j \sigma(w_j x + b_j)$$

$$\mathcal{F}_\theta = \{f_\theta : \theta \in \Theta\}$$

Let $\mathcal{G} = \{(x, y) \mapsto \ell(x, f(y)), f \in \mathcal{F}_\theta\}$

Proposition: Assume that ℓ is G -Lipschitz and $\sigma = \text{ReLU}$, then

$$\Theta = \left\{ \vec{a}, \vec{w}, \vec{b} : \begin{cases} \|w_j\|^2 + \frac{b_j^2}{R^2} \leq 1 \quad \forall j \\ \|a\|_1 \leq D \end{cases} \right\}$$

$$\text{Rad}_n(\mathcal{G}) \leq \frac{4GD R}{\sqrt{n}}$$

Lemma: Let φ_j be G -Lipschitz cont, and $\varepsilon_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif} \{-1, 1\}$:

$$\mathbb{E}_\varepsilon \sup_{\theta} \sum_{j=1} \varepsilon_j \varphi_j(a_j(\theta) + b_j(\theta)) \leq G \mathbb{E}_\varepsilon \sup_{\theta} \sum_{j=1} \varepsilon_j a_j(\theta) + b_j(\theta)$$

Proof:

$$\begin{aligned} \mathbb{E}(\mathcal{R}(\hat{f}) - \inf_{f \in \mathcal{F}} \mathcal{R}(f)) &\leq \mathbb{E}(\mathcal{R}(\hat{f}) - \hat{\mathcal{R}}(\hat{f}) + \underbrace{\hat{\mathcal{R}}(\hat{f}) - \hat{\mathcal{R}}(g_{\mathcal{F}})}_{\leq 0} + \hat{\mathcal{R}}(g_{\mathcal{F}}) - \mathcal{R}(g_{\mathcal{F}})) \\ &\leq 2 \mathbb{E}(\sup_{f \in \mathcal{F}} (\hat{\mathcal{R}}(f) - \mathcal{R}(f))) \leq 4 \text{Rad}_n(\mathcal{G}). \end{aligned}$$

we also have, by the above Lemma

$$\begin{aligned} \text{Rad}_n(\mathcal{G}) &\leq G \text{Rad}_n(\mathcal{F}_{\Theta}) = G \mathbb{E}_{\mathcal{E}, \mathcal{D}} \left(\sup_{\Theta} \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m \varepsilon_i a_j \sigma(w_j x_i + b_j) \right) \\ &\xrightarrow{\text{sup over } \|\vec{a}\| \leq D} \leq G \mathbb{E}_{\mathcal{E}, \mathcal{D}} \left(\sup_{\Theta} \sup_{\substack{\|w_j\|_1 + \frac{|b_j|^2}{R^2} \leq 1}} D \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i \sigma(w_j x_i + b_j) \right| \right) \end{aligned}$$

$$\xrightarrow{\text{Lipshitz } \sigma} \leq 2GD \mathbb{E}_{\mathcal{E}, \mathcal{D}} \left(\sup_{\Theta} \sup_{w, b} \left| w_j^T \left(\frac{1}{n} \sum \varepsilon_i x_i \right) + b_j \left(\frac{1}{n} \sum \varepsilon_i \right) \right| \right)$$

$$\begin{aligned} &\xrightarrow{\text{C.S. + optimize over } w, b} \leq 2GD \mathbb{E}_{\mathcal{E}, \mathcal{D}} \left(\sqrt{\left(\frac{1}{n} \sum \varepsilon_i x_i \right)^2 + R^2 \left(\frac{1}{n} \sum \varepsilon_i \right)^2} \right) \\ &\leq 2GD \sqrt{\mathbb{E}_{\mathcal{E}, \mathcal{D}} \left(\frac{1}{n} \sum \varepsilon_i x_i \right)^2 + R^2 \mathbb{E}_{\mathcal{E}} \left(\frac{1}{n} \sum \varepsilon_i \right)^2} \\ &\leq 2GD \sqrt{\frac{1}{n} \mathbb{E}_{\mathcal{D}}(\|x\|_2^2) + \frac{1}{n} R^2} \leq \frac{2\sqrt{2} GDR}{\sqrt{n}} \end{aligned}$$

There are many extensions of the above for multilayer networks, e.g.

Thm: Let the activations σ_j be L -Lipschitz, and assume that

$$\|W_{\ell}^T\|_{1, \infty} = \sup_{\|v_{\ell}\|_1 \leq 1} \|W_{\ell}^T v_{\ell}\|_{\infty} \leq B$$

$$\text{then } \text{Rad}(\mathcal{F}_{\Theta}, S) \leq \|X^T\|_{2, \infty} (2pB)^{L+1} \sqrt{2p \log d}$$

where $\|X\|_{2,\infty}^2 = \sum_{j=1}^n \|x_j\|_\infty^2 \leq n$

Proof (♦) Let $\mathcal{D}' = \{z'_1, \dots, z'_n\}$ be an independent copy of the data $\mathcal{D} = \{z_1, \dots, z_n\}$. Let $(\varepsilon_i)_{i \in \{1, \dots, n\}}$ be i.i.d. Rademacher random variables, which are also independent of $\mathcal{D}$ and $\mathcal{D}'$. Using that for all i in $\{1, \dots, n\}$, $\mathbb{E}[h(z'_i)|\mathcal{D}] = \mathbb{E}[h(z)]$, we have:

$$\begin{aligned} \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\mathbb{E}[h(z)] - \frac{1}{n} \sum_{i=1}^n h(z_i) \right) \right] &= \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{E}[h(z'_i)|\mathcal{D}] - \frac{1}{n} \sum_{i=1}^n h(z_i) \right) \right] \\ &= \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{E}[h(z'_i) - h(z_i)|\mathcal{D}] \right) \right], \end{aligned}$$

by definition of the independent copy $\mathcal{D}'$. Then

$$\mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\mathbb{E}[h(z)] - \frac{1}{n} \sum_{i=1}^n h(z_i) \right) \right] \leq \mathbb{E} \left[\mathbb{E} \left(\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n [h(z'_i) - h(z_i)] \right) \middle| \mathcal{D} \right) \right],$$

using that the supremum of the expectation is less than the expectation of the supremum. Thus, by the tower law of expectation, we get

$$\mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\mathbb{E}[h(z)] - \frac{1}{n} \sum_{i=1}^n h(z_i) \right) \right] \leq \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n [h(z'_i) - h(z_i)] \right) \right].$$

We can now use the symmetry of the laws of ε_i and $h(z'_i) - h(z_i)$, to get:

$$\begin{aligned} &\mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\mathbb{E}[h(z)] - \frac{1}{n} \sum_{i=1}^n h(z_i) \right) \right] \\ &\leq \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i (h(z'_i) - h(z_i)) \right) \right] \\ &\leq \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i (h(z_i)) \right) \right] + \mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i (-h(z_i)) \right) \right] \\ &= 2\mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \varepsilon_i h(z_i) \right) \right] = 2R_n(\mathcal{H}). \end{aligned}$$

The reasoning is essentially identical for $\mathbb{E} \left[\sup_{h \in \mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n h(z_i) - \mathbb{E}[h(z)] \right) \right] \leq 2R_n(\mathcal{H})$.

■

Summary: choose $p \gg 1$, $u \gg p$, convex

